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ROP-VL: An Open Dataset Integrating Fundus Photograph and Clinical Knowledge for Retinopathy of Prematurity

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ROP-VL: An Open Dataset Integrating Fundus Photograph and Clinical Knowledge for Retinopathy of Prematurity

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The development of advanced artificial intelligence (AI) for Retinopathy of Prematurity (ROP) is hindered by a scarcity of public datasets that integrate color fundus photographs with the expert clinical knowledge required for building robust vision-language models. We present ROP-VL, an open-access dataset designed to bridge this critical gap. It contains 2,020 color fundus photographs from 1,116 infants, encompassing eight diagnostic categories including Normal, A-ROP, Laser-treated ROP, and ROP Stages 1-5. ROP-VL explicitly pairs these images with patient biological metadata and structured textual descriptions, providing the necessary components for multimodal learning. We perform a technical validation by training four image-only models and a vision-language model on ROP occurrence and severity classification tasks. The results demonstrate that the vision-language model, leveraging the integrated structured clinical data, consistently achieves superior performance. ROP-VL provides a purpose- built resource to facilitate the development of more robust, knowledge-aware AI, aiming to advance automated ROP diagnosis.

Datasets:

Repository Name	Dataset Title	Accession Number or DOI	URL to data record	Private reviewer access URL/code
ROP-VL: An Open Dataset Integrating Fundus Photograph and Clinical Knowledge for Retinopathy of Prematurity	ROP-VL: An Open Dataset Integrating Fundus Photograph and Clinical Knowledge for Retinopathy of Prematurity	10.6084/m9.figshare.30143461	https://doi.org/10.6084/m9.figshare.30143461.v1	

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ROP-VL: An Open Dataset Integrating Fundus Photograph and Clinical Knowledge for Retinopathy of Prematurity

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Abstract

The development of advanced artificial intelligence (AI) for Retinopathy of Prematurity (ROP) is hindered by a scarcity of public datasets that integrate color fundus photographs with the expert clinical knowledge required for building robust vision-language models. We present ROP-VL, an open-access dataset designed to bridge this critical gap. It contains 2,020 color fundus photographs from 1,116 infants, encompassing eight diagnostic categories including Normal, A-ROP, Laser-treated ROP, and ROP Stages 1-5. ROP-VL explicitly pairs these images with patient biological metadata and structured textual descriptions, providing the necessary components for multimodal learning. We perform a technical validation by training four image-only models and a vision-language model on ROP occurrence and severity classification tasks. The results demonstrate that the vision-language model, leveraging the integrated structured clinical data, consistently achieves superior performance. ROP-VL provides a purpose-built resource to facilitate the development of more robust, knowledge-aware AI, aiming to advance automated ROP diagnosis.

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Background & Summary

The reduction of childhood blindness represents a primary objective of the World Health Organization's vision programs [1, 2]. Retinopathy of Prematurity (ROP) stands as a leading cause of visual impairment and irreversible blindness in children worldwide [3, 4]. It is a vasoproliferative disease that primarily affects premature infants of very low birth weight or gestational age. Early screening for ROP is of paramount importance for this vulnerable population. The early detection and appropriate treatment of ROP can significantly prevent adverse visual outcomes and reduce associated healthcare costs [5]. In accordance with the International Classification of Retinopathy of Prematurity (ICROP) [6, 7], the disease is staged by severity from Stage 1 (a thin demarcation line at the vascular-avascular junction) to Stage 5 (total retinal detachment). An aggressive, rapidly progressing form, known as Aggressive-ROP (A-ROP), also exists. While most cases of ROP are mild and regress spontaneously, timely and appropriate intervention for infants identified as requiring treatment is essential to prevent retinal detachment and subsequent blindness [8].

The current standard for ROP diagnosis relies on subjective assessment by experienced ophthalmologists using indirect ophthalmoscopy. The clinical diagnosis of ROP remains subjective, and significant inter-observer variability is documented even among experts [9, 10]. Because ROP screening is a specialized field that yields digital images, it represents an ideal domain for the application of computer-aided diagnosis. Artificial intelligence (AI) is increasingly applied to the automated diagnosis of various retinal diseases [11–20]. AI technologies offer a scalable means to extend expert-level medical services to remote or underserved areas, thereby accelerating the diagnostic process and ensuring more patients receive timely treatment. Early research often employs traditional regression analysis based on birth characteristics (e.g., birth weight, gestational age, delivery mode, and gender) to predict ROP occurrence [13, 21, 22]. However, these models have inherent limitations as they often neglect the critical state of the retina itself [23].

More recently, a growing body of AI research focuses on analyzing color fundus photographs (CFPs) to identify clinical features of ROP [24–27], including its presence, severity, and the existence of plus disease. These studies underscore the importance and suitability of CFPs for assessing retinal diseases and have led to AI systems capable of automated ROP identification and diagnosis. However, these image-only models do not leverage the rich, contextual information contained in clinical reports and expert knowledge [28, 29]. The integration of textual data, which describes the specific pathologies and patient context, with visual features from CFPs is crucial for building more robust and interpretable models [30]. This fusion of modalities, facilitated by vision-language models, mirrors the diagnostic process of a human expert who considers both the image and the patient's history, holding the potential to improve diagnostic accuracy and provide more nuanced, knowledge-aware predictions. Despite these advances, few systems meet the rigorous standards required for real-world clinical application, indicating a need for further research. A significant barrier to progress is the scarcity of public datasets. The majority of large, freely available retinal datasets focus on adult pathologies such as diabetic retinopathy, glaucoma, or pathological

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myopia [31–33]. In contrast, resources for infant retinal imaging, and for ROP specifically, remain critically lacking. The few publicly available ROP datasets that do exist often suffer from limitations [34, 35], such as an incomplete range of diagnostic categories or a lack of integrated domain knowledge and biological metadata. Therefore, the development of an ideal CFP dataset for infants, complete with matched clinical knowledge, is essential to bridge this gap in clinical AI research.

Table 1 compares ROP-VL with the key existing publicly available ROP datasets. ROP-VL is distinguished by its comprehensive diagnostic coverage of all eight clinical categories (including all five ROP stages, A-ROP, and Laser-treated ROP), its explicit provision of structured expert clinical knowledge as textual descriptions for each category, rich patient biological metadata (five fields per patient), and its purpose-built one-to-one text-image pairing via $\pi_{\text{ROP-EK}}(\cdot)$. To our knowledge, ROP-VL is the first publicly available ROP dataset designed specifically to support multimodal vision–language model development.

Table 1 Comparison of publicly available ROP fundus image datasets.

Dataset	Images	Metadata	Knowledge	MM Pairing
Czech-Ostrava [35]	6,004	Partial	No	No
Sci-Data 2024 [34]	1,099	No	No	No
ROP-VL (Ours) [36]	**2,020**	**Yes**	**Yes**	**Yes**

Metadata: patient biological metadata (gender, gestational age, birth weight, plurality, delivery mode). Knowledge: structured expert clinical descriptions per category. MM Pairing: multimodal text-image pairing.

To address these limitations and facilitate future research, we present ROP-VL [36], an open-access database containing 2,020 CFPs from 1,116 subjects, paired with corresponding biological metadata and curated domain knowledge for each disease. The dataset encompasses a comprehensive range of categories, including Normal, A-ROP, Laser-treated ROP, and all five stages of ROP (Stage 1 to Stage 5). A schematic overview of this study is presented in Fig. 1. The primary importance of this dataset lies in its multimodal structure. By explicitly linking high-quality images with structured textual descriptions of both the disease and the patient's context, ROP-VL is purpose-built to support the development of next-generation, knowledge-aware vision-language models that can more closely emulate clinical reasoning. We anticipate that this dataset will provide a valuable resource for developing models with greater clinical applicability. Consequently, the release of the ROP-VL dataset is intended to provide additional impetus for the development of new AI models and the validation of existing ones. We hope the release of ROP-VL will aid researchers globally, accelerating the advancement of computer-aided systems for ROP diagnosis and treatment.

The multimodal structure of ROP-VL makes it well-suited for a broad range of vision–language tasks beyond standard image classification. The one-to-one text-image pairs are directly compatible with CLIP-style contrastive pretraining [29], where the expert knowledge text serves as the positive anchor for its paired fundus image. The structured clinical descriptions can be reformatted as instruction-answer pairs for instruction fine-tuning of large vision–language models such as LLaVA. Furthermore,

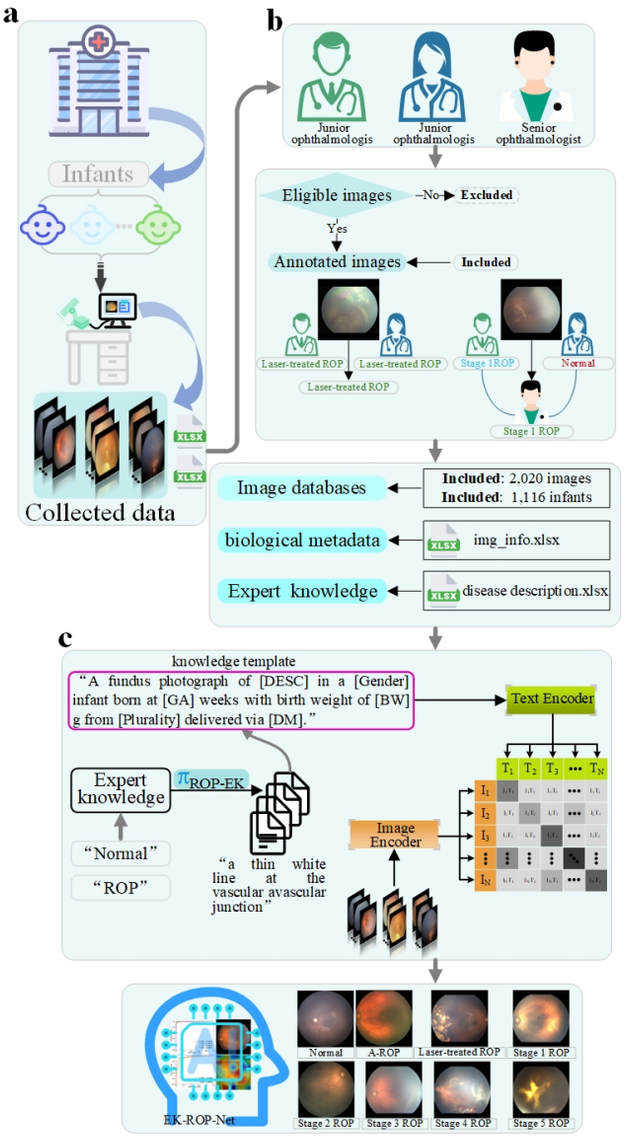


Fig. 1 Workflow for dataset construction and development of the EK-ROP-Net model. a. Clinical data acquisition. b. Image annotation by ROP specialists. c. Construction and performance validation of the EK-ROP-Net automated screening model.

the dataset supports text-image retrieval (enabling natural language queries for clinical decision support), clinical description generation (automating report writing from fundus images), and visual question answering (VQA). A trained dual-encoder model can also leverage the eight category descriptions as fixed class prototypes for zero-shot or few-shot language-conditioned classification.

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Methods

Data collection. The data in the ROP-VL dataset come from routine examinations conducted at the Shenzhen Eye Hospital (SZEH), China, between 2005 and 2023. The screening program primarily targets all infants born before 32 weeks of gestational age or with a birth weight below 2,000 grams. However, older or heavier infants are also included in the screening based on their clinical condition, particularly those who are clinically unstable or require supportive care such as oxygen therapy. Prior to the examination, parents or legal guardians of the infants provide written informed consent for the use of the data in scientific research. All associated images and records are fully anonymized to protect patient privacy. This study adheres to the principles of the Declaration of Helsinki and operates under the approval of the Medical Ethics Committee of SZEH (ID: 2023KYPJ091). The CFPs in this study are acquired using three wide-field contact fundus imaging systems: the RetCam (Clarity Medical Systems, Inc., USA), the ROP Screening System (SW-8000, China), and the Nautilus Fundus Imaging System (RS-B002, China). For each examination, infant pupils are dilated using tropicamide approximately 30 minutes prior. Fundus photography is then performed by well-trained technicians or ophthalmologists with the assistance of several staff members.

The inclusion criteria for CFPs are as follows: (1) Absence of significant quality issues, such as artifacts, defocus, motion blur, or improper exposure, that compromise the clarity of the target region. (2) For normal images, the focus is sharp on the retina. (3) For images containing pathology, the focus is clearly centered on the lesion area. (4) Clear visibility of anatomical structures and pathologies, including the retina, vasculature, laser scars, and ROP lesions. Overall, image quality is a critical evaluation standard in our screening process. Images that are redundant, of poor quality, contain unclear lesion depictions, or have missing source information are excluded. All images included in this study are considered clinically acceptable and usable, featuring clear views of the retina, vessels, and pathologies. The selected images represent the standard six-orientation, primarily including views centered on the optic disc, macula, and positions temporal, nasal, superior, and inferior to the optic disc.

Image quality assessment was performed independently by two qualified ophthalmologists with experience in retinal imaging. The specific evaluation criteria included: (i) absence of significant defocus or motion blur; (ii) absence of improper exposure (over- or underexposure) or strong light reflections obscuring the region of interest; (iii) adequate visibility of retinal anatomical structures (optic disc, blood vessels, avascular zone); and (iv) clear visualization of pathologically relevant features (e.g., demarcation lines, neovascularization, laser scars). Quality assessment was conducted through clinical visual inspection rather than automated quantitative metrics (e.g., signal-to-noise ratio or Laplacian-variance blur detection), reflecting standard clinical practice in which fundus image suitability is inherently context-dependent. When the two ophthalmologists disagreed on the acceptability of a borderline-quality image, the case was escalated to a third, more senior ophthalmologist, whose judgment was final. This three-stage escalation protocol is consistent with the diagnostic labeling procedure described under “Disease diagnosis.”

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The final dataset comprises 2,020 CFPs from 1,116 subjects. A minimum of one eye is imaged per subject, and only one high-quality CFP per eye is retained for the dataset. Due to the progressive nature of ROP, the data include images from multiple follow-up visits for the same infant. Furthermore, the dataset includes some infants with severe ROP who have undergone laser treatment. Among the 1116 subjects, 63.5% are male. The mean gestational age is 30.43 ± 3.87 weeks, and the mean birth weight is 1504.75 ± 782.33 g. All subjects in the ROP-VL dataset are of Chinese ethnicity. As the images captured by different devices vary in size, a pre-processing technique is implemented. We first crop the black background from each image. The cropped region is then centered on a new black square canvas. This process standardizes the image dimensions to a uniform square format while preserving the aspect ratio of the retinal area, thus ensuring that the image content is not distorted. In addition to the CFP data, this study also provides image information and the patients' biological metadata. Further details on the dataset and subjects are provided in the Section 2 Data Records.

Disease diagnosis. The image classification in this study strictly adheres to the ICROP [6, 7] guidelines. The dataset incorporates eight distinct categories: Normal, Laser-treated ROP, A-ROP, Stage 1 ROP, Stage 2 ROP, Stage 3 ROP, Stage 4 ROP, and Stage 5 ROP. The distribution of data across these categories is detailed in Table 2, and representative fundus images for each class are presented in Fig. 2. The diagnostic labels are assigned by three experienced ophthalmologists from SZEH. Specifically, two junior ophthalmologists independently classify all fundus images. In cases where their classifications are inconsistent, a third, more senior ophthalmologist reviews the disputed images and makes the final determination.

Table 2 Distribution of patients and images for given diagnosis.

Diagnosis	Patient	Image	Image Percentage	Patient Percentage
Normal	381	623	30.8%	34.1%
Laser-treated ROP	224	345	17.1%	20.1%
A-ROP	103	148	7.3%	9.2%
Stage 1 ROP	180	236	11.7%	16.1%
Stage 2 ROP	253	338	16.7%	22.7%
Stage 3 ROP	174	241	11.9%	15.6%
Stage 4 ROP	46	59	2.9%	4.1%
Stage 5 ROP	27	30	1.5%	2.4%
Total	1,116	2,020	100%	100%

Note: Patient total count is deduplicated to account for follow-up data.

To intuitively observe the feature distribution of the eight different categories within the dataset, we employ the t-distributed Stochastic Neighbor Embedding (t-SNE) technique [37]. t-SNE is a nonlinear dimensionality reduction algorithm primarily used for visualizing high-dimensional data in a low-dimensional space, such as a two-dimensional map. It models the similarity between data points in the high-dimensional feature space as a probability distribution and then optimizes a similar

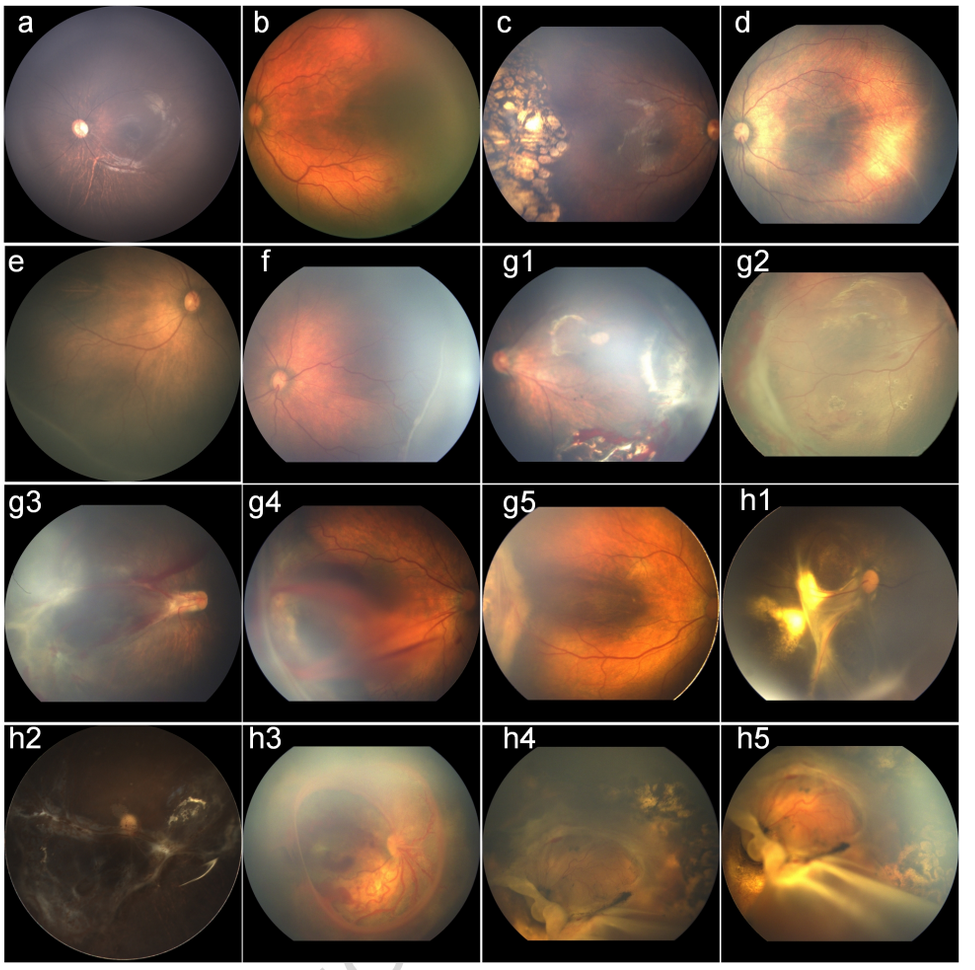


Fig. 2 Demonstration of various kinds of color fundus photography. a. normal image; b. A-ROP; c. Laser-treated ROP; d. Stage 1 ROP; e. Stage 2 ROP; f. Stage 3 ROP; g1-g5. Stage 4 ROP (partial retinal detachment with visible traction); h1-h5. Stage 5 ROP (total funnel-shaped retinal detachment).

distribution in the low-dimensional map. This process arranges the data points so that similar images are modeled by nearby points and dissimilar images are modeled by distant points, thereby revealing the underlying structure of the data. The result is shown in Fig. 3. The resulting visualization reveals a complex distribution with considerable overlap among the different diagnostic categories. This intermingling of clusters is clinically plausible and reflects the continuous nature of ROP progression, where the visual distinctions between adjacent stages (e.g., Stage 1 ROP and Stage 2 ROP) can be subtle. Furthermore, the high intra-class variability of the disease's presentation contributes to the dispersion of data points. Therefore, the t-SNE plot does not suggest flawed feature extraction but rather highlights the inherent difficulty of

the classification task. The lack of clear separation in this 2D visualization underscores the necessity for developing sophisticated, high-dimensional models that can learn the nuanced and overlapping features required for an accurate diagnosis.

Several inherent limitations of t-SNE should be noted when interpreting this visualization: (1) **Stochasticity:** t-SNE is a stochastic algorithm and results may vary between runs. The figure represents one specific realization using a fixed random seed. (2) **Non-preservation of global distances:** Inter-cluster distances in the 2D projection do not accurately reflect distances in the original high-dimensional feature space. The visualization should not be used to make quantitative inferences about inter-class similarity. (3) **Crowding effect:** t-SNE tends to compress distant points together in low dimensions, which may artificially increase the apparent overlap between categories. Despite these limitations, the visualization effectively reveals the relative feature distribution of the eight diagnostic categories and the classification challenge inherent in ROP staging.

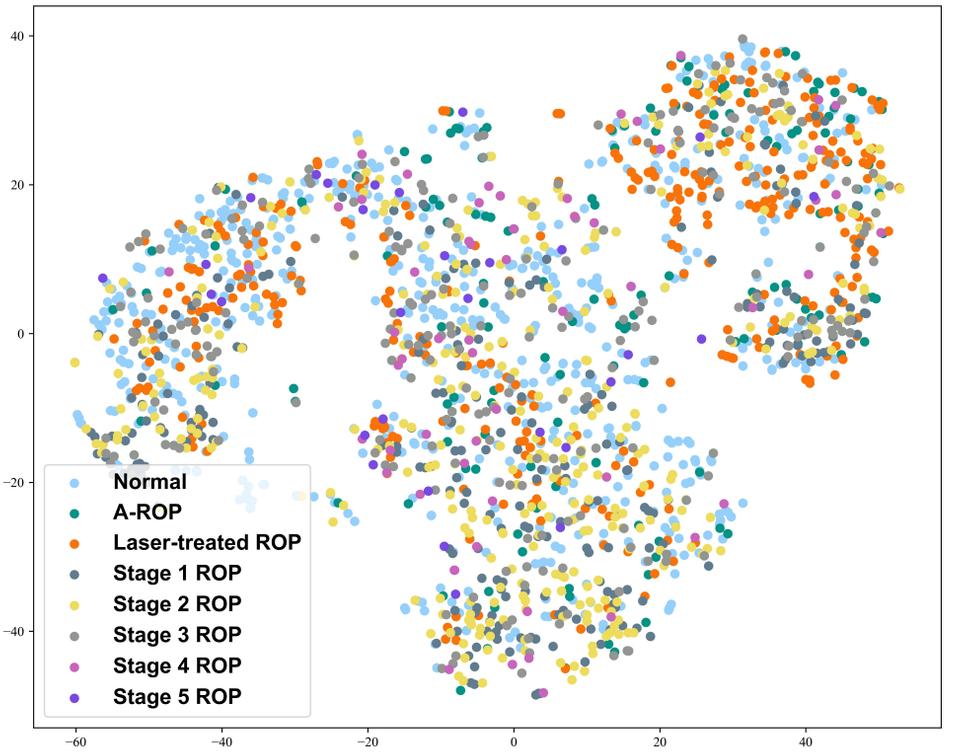


Fig. 3 t-SNE visualization of the ROP-VL dataset. Each point represents the high-dimensional features of a single fundus image, projected into a two-dimensional space. Points are colored according to their respective diagnostic category. The considerable overlap between categories reflects the continuous nature of ROP progression and the inherent challenge of fine-grained staging.

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ROP's domain knowledge construction. In this work, we construct a descriptor for ROP expert knowledge (ROP-EK), denoted as $\pi_{ROP-EK}(\cdot)$. This transformation maps each patient's case, defined by their diagnosis and metadata, to a textual description that encapsulates both the relevant clinical findings and the patient's biological metadata. This description is systematically generated using the following structured template: "A fundus photograph of [DESC] in a [Gender] infant born at [GA] weeks with birth weight of [BW] g from [Plurality] delivered via [DM]." Within this template, the placeholders are populated as follows: [DESC] represents the clinical description of the disease. This text is retrieved from the 'disease_description.xlsx' file, corresponding to the diagnostic category of the CFP. [GA] (Gestational Age), [BW] (Birth Weight), [DM] (Delivery Mode), [Gender], and [Plurality] constitute the patient's biological metadata, which are obtained from the 'img_info.xlsx' file. This form of expert domain knowledge is often overlooked in traditional training pipelines. Its integration with CFPs as a textual description provides a crucial component for building powerful and knowledge-aware vision-language models.

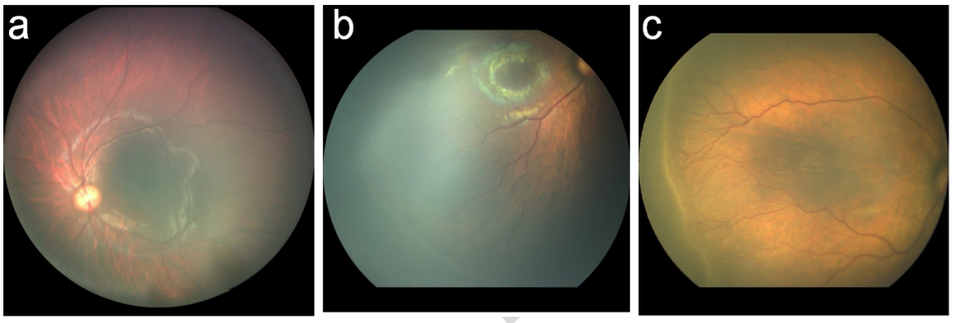


Fig. 4 Three representative text–image pairing examples. a. "A fundus photograph of healthy, no findings, no lesion signs, no retinopathy in a male infant born at 40 weeks with birth weight of 2880g from singleton pregnancy delivered via cesarean section." b. "A fundus photograph of a thin white line at the vascular-avascular junction with possible abnormal vessel branching in a female infant born at 27 weeks with birth weight of 1000g from singleton pregnancy delivered via vaginally." c. "A fundus photograph of ridge with width and height that evolves from the demarcation line, the ridge is raised and thicker than stage 1 ROP in a male infant born at 33 weeks with birth weight of 1480g from singleton pregnancy delivered cesarean section."

Annotation workflow and validation. The clinical descriptions for the eight diagnostic categories were initially drafted by two experienced ROP ophthalmologists (each with $\geq$5 years of dedicated ROP clinical experience) from SZEH, based on the cardinal visual features defined in the ICROP guidelines [6, 7]. The descriptions focus exclusively on features directly observable in color fundus photographs (e.g., morphology of demarcation lines, characteristics of neovascularization, appearance of laser scars). The drafted descriptions were subsequently reviewed and validated by a senior ophthalmologist ($\geq$15 years of ROP diagnosis experience), who confirmed their clinical accuracy and ICROP alignment. Because the descriptions are derived from well-established ICROP guidelines and represent objective, visually observable features, formal quantitative inter-rater agreement metrics (e.g., Cohen's kappa) were not applied to the

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description-drafting process. All discrepancies between the two drafting specialists were resolved through consensus discussion with the senior reviewer. The descriptions vary systematically across disease stages, reflecting the progressive pathological severity: from a thin demarcation line (Stage 1) to a raised ridge with neovascular tufts (Stage 2), to extraretinal proliferation (Stage 3), and finally to partial (Stage 4) and total (Stage 5) retinal detachment, ensuring clinically meaningful and discriminative textual representations. To illustrate the complete output of $\pi_{\text{ROP-EK}}(\cdot)$, three representative text–image pairing examples are provided in Fig. 4

Data Records

The ROP-VL dataset is publicly available on Figshare [36] and includes both an image folder and a data summary in XLSX format. To ensure patient anonymity, the dataset contains no personally identifiable information. The dataset is provided as a single zipped file containing eight subfolders, each of which corresponds to a specific diagnostic category: Normal, Laser-treated ROP, A-ROP, Stage 1 ROP, Stage 2 ROP, Stage 3 ROP, Stage 4 ROP, and Stage 5 ROP. Each subfolder contains all CFPs that belong to its respective class. In addition, two Excel files, 'img_info.xlsx' and 'disease_description.xlsx', are included, as Fig. 5 illustrates.

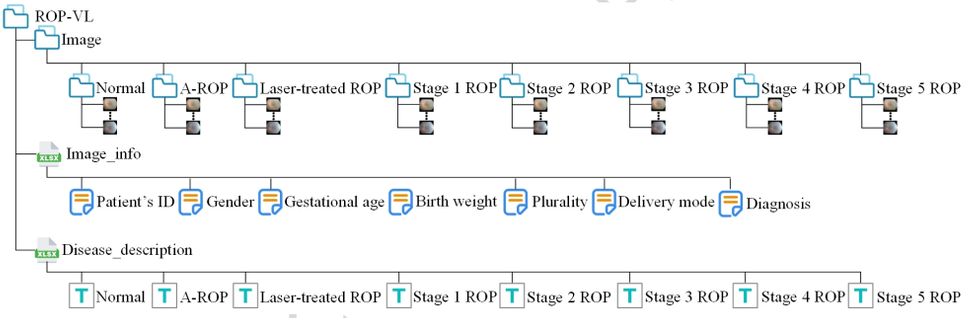


Fig. 5 Folder organization of our ROP-VL dataset.

The CFPs within each subfolder are stored in JPEG format with 8 bits per color channel. The 'img_info.xlsx' file contains detailed information for each image and the corresponding patient's biological metadata. The following sections describe the dataset parameters and variables, which are also summarized in Table 3.

Patient's ID. The patient ID is a fully anonymized integer that serves as a unique identifier. Its processing ensures that no one, not even the patients themselves, can link the images and data back to a specific individual.

Gender. This variable indicates the sex of the patient, with possible values of 'male' or 'female'. The dataset comprises images from 1116 patients: 709 males and 407 females. In total, there are 1278 retinal images from male patients and 742 from female patients.

Gestational age. This integer value represents the patient's gestational age in weeks at the time of birth. The mean gestational age is 30.40 weeks (standard deviation

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Table 3 Data dictionary: field definitions, data types, accepted value ranges/units, and missing data coding.

Parameter	Data type	Possible Values / Range	Unit	Column name
Image name	String	–	–	image
Patient's ID	Integer	–	–	patient_id
Gender	String	male / female	–	gender
Gestational age	Integer	23–42	weeks	gestational_age
Birth weight	Integer	450–5,400	grams	birth_weight
Plurality	Integer	1 = singleton, 2 = twin, 3 = triplet	–	plurality
Delivery mode	String	vaginally / cesarean section	–	delivery_mode
Diagnosis	String	Normal / Laser-treated ROP / A-ROP / Stage 1 ROP / Stage 2 ROP / Stage 3 ROP / Stage 4 ROP / Stage 5 ROP	–	categories

Note: All fields are complete; no missing values exist in the dataset.

(SD) = 3.85). The minimum gestational age in the dataset is 23 weeks, and the data include 11 patients with a gestational age of 40 weeks or more. The distribution of patients and retinal images by gestational age is shown in Fig. 6.

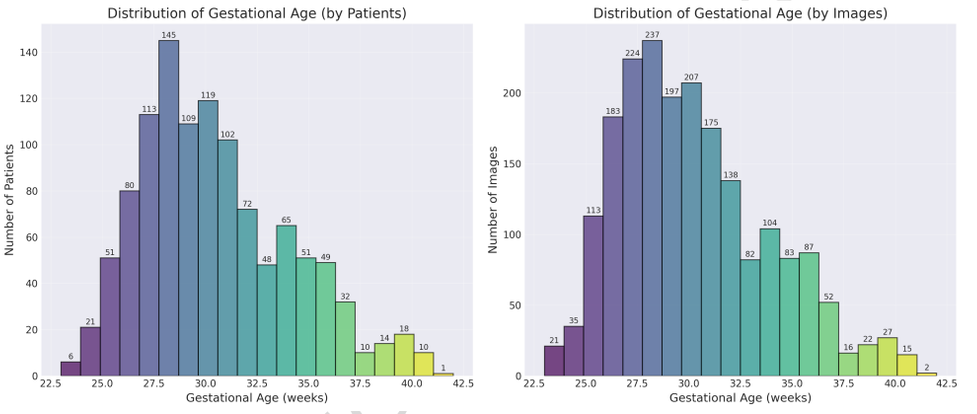


Fig. 6 Distribution of patients and retinal images across gestational ages (weeks).

Birth weight. This variable records the patient's weight in grams at birth. The mean birth weight is 1490.37 g (SD = 732.03 g), with a minimum of 450 g and a maximum of 5400 g. The distribution of patients and retinal images by birth weight is depicted in Fig. 7.

Plurality. This integer value indicates whether a patient is from a singleton or multiple gestation birth. The dataset includes 801 patients from singleton births and 315 from multiple births, which correspond to 1444 and 576 retinal images, respectively.

Delivery mode. This variable specifies the method of birth, with values of 'vaginally' or 'caesarean section'. The dataset contains records for 589 patients delivered vaginally and 527 via caesarean section. This corresponds to 1093 retinal images from vaginal births and 927 from caesarean section births.

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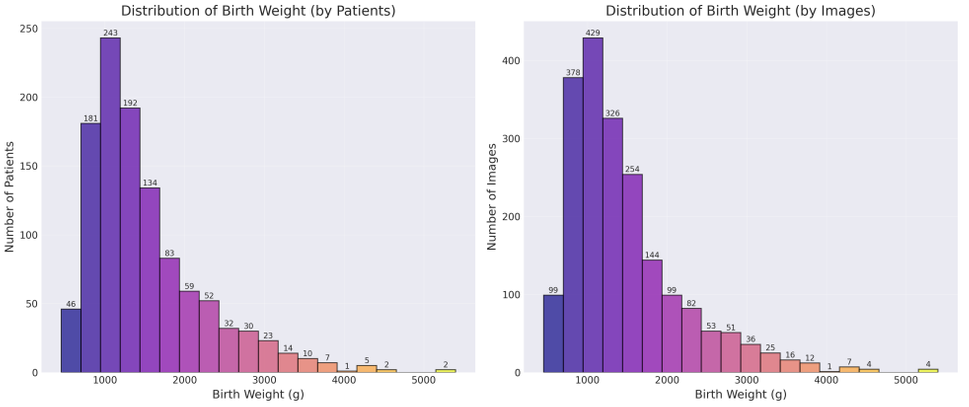


Fig. 7 Distribution of patients and retinal images across birth weight (grams).

Diagnosis. This string variable details the clinical diagnosis for each image. The dataset encompasses eight defined diagnostic categories. The complete list of possible diagnoses and their distribution is provided in Table 2.

The 'disease_description.xlsx' file contains curated clinical descriptions for the eight fundus conditions included in the ROP-VL dataset. This textual data is compiled in accordance with the ICROP and has been validated by expert ophthalmologists.

Technical Validation

To assess the utility of the ROP-VL dataset for training and testing AI models, we develop five deep learning models. First, we implement four image-only AI models to automatically identify ROP-related conditions from CFPs: ResNet50 [38], ConvNeXt-T [39], Swin-T [40], and EfficientVim-M [41]. Further, we integrate the CFPs with the ROP expert knowledge to develop a more robust vision-language model for ROP diagnosis, which we refer to as EK-ROP-Net. This model uses a ResNet50 as its vision encoder and BioClinicalBERT [42] as its text encoder. We evaluate these models on two distinct classification tasks using the ROP-VL dataset: ROP Occurrence (Four-Class Task): This task involves the automatic identification of 'Normal', 'Any Stage of ROP' (grouping Stage 1 to 5), 'A-ROP', and 'Laser-treated ROP'. ROP Severity (Three-Class Task): This task focuses on automatically classifying disease severity into 'Mild ROP' (Stage 1 and Stage 2), 'Moderate ROP' (Stage 3), and 'Severe ROP' (Stage 4 and Stage 5).

In our experiments, we employ a five-fold cross-validation scheme where the ROP-VL dataset is partitioned at the patient level. This strategy ensures that all images from a single subject are exclusively contained within the same fold in any given iteration. By splitting the data this way, we prevent data leakage and guarantee that the models learn generalizable pathological features rather than patient-specific characteristics, leading to a more reliable and unbiased evaluation of performance. The patient-level partition is performed first, and data augmentation is applied exclusively to the training split of each fold during model training. The validation set receives no

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augmentation. Exact image and subject counts for each fold are provided in Table 4. All models are implemented using the PyTorch framework and are trained on an NVIDIA A40 GPU. Prior to training, all input images are resized to 512 × 512 pixels. During the training process, training images are first intensity-normalized to be in the [0, 1] range and then augmented using random horizontal flipping, random rotation within a [-5, 5] degree range, scaling within a [0.9, 1.1] range, and color jittering. Validation images are only resized and intensity-normalized (no augmentation). The models are optimized using the Adam optimizer. The initial learning rate is set to 0.0001 and is managed by a dynamic schedule combined with cosine annealing to ensure stable and effective convergence. Each model is trained for 100 epochs with a batch size of 128. We report model performance on the separate validation set for each fold of the cross-validation experiment.

Table 4 Five-fold cross-validation split summary.

Disease	Train subjects	Train images	Valid subjects	Valid images
Normal	305/304/306/304/305	501/505/494/490/502	76/77/75/77/76	122/118/129/133/121
Laser-treated ROP	179/179/180/179/179	277/278/281/270/274	45/45/44/45/45	68/67/64/75/71
A-ROP	82/82/82/83/83	120/117/117/119/119	21/21/21/20/20	28/31/31/29/29
Stage 1 ROP	144/144/144/144/144	188/184/188/192/192	36/36/36/36/36	48/52/48/44/44
Stage 2 ROP	202/202/203/202/203	266/272/271/273/270	51/51/50/51/50	72/66/67/65/68
Stage 3 ROP	138/139/139/140/140	190/191/192/200/191	36/35/35/34/34	51/50/49/41/50
Stage 4 ROP	36/36/37/37/38	45/46/48/47/50	10/10/9/9/8	14/13/11/12/9
Stage 5 ROP	21/21/22/22/22	22/24/25/24/25	6/6/5/5/5	8/6/5/6/5

EK-ROP-Net inference procedure. EK-ROP-Net is trained using a contrastive learning framework analogous to CLIP. During training, each image is paired with its corresponding expert knowledge text generated by $\pi_{\text{ROP-EK}}(\cdot)$, and the model learns to align the visual and textual representations of ROP pathology through contrastive loss. At inference time, only the visual encoder is used: the features extracted from the test image by the trained visual encoder are directly used for classification, without requiring the text encoder or any disease description input. Since disease-specific knowledge texts are entirely absent from the inference pipeline, no label information is accessible to the model at test time, ensuring an unbiased evaluation. The performance advantage of EK-ROP-Net over the image-only baselines therefore reflects the improved visual representations acquired through contrastive training with domain knowledge texts, rather than any direct use of class descriptions at inference time. As an alternative training configuration, researchers may train a metadata-only variant by substituting the disease description component [DESC] in the $\pi_{\text{ROP-EK}}(\cdot)$ template with a generic placeholder during training, thereby isolating the contribution of clinical descriptions from patient metadata.

The quantitative results of the model evaluations are presented in Table 5 for the ROP Occurrence task and Table 6 for the ROP Severity task. These findings are further detailed in Fig. 8 and Fig. 9, which respectively display the confusion matrices, Receiver Operating Characteristic (ROC) curves, and t-SNE visualizations for each task. Across both classification tasks, the experimental results demonstrate the value of the ROP-VL dataset for developing effective diagnostic models. For the

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Table 5 Model classification performances in ROP Occurrence task (%).

Method	Acc	AUC	Pre	Sen	F1	Spec	Kappa
ResNet50	92.83 ± 1.12	98.03 ± 0.60	92.05 ± 1.32	87.19 ± 2.10	89.07 ± 1.39	97.19 ± 0.48	89.18 ± 1.68
ConvNeXt-T	92.63 ± 1.19	98.97 ± 0.34	92.33 ± 0.62	89.16 ± 1.82	90.32 ± 1.27	97.27 ± 0.52	88.98 ± 1.86
Swin-T	92.77 ± 0.26	98.86 ± 0.26	91.83 ± 1.22	88.82 ± 2.96	89.77 ± 1.73	97.25 ± 0.24	89.16 ± 0.48
EfficientViM-M3	90.85 ± 0.99	97.76 ± 0.51	90.28 ± 2.46	85.80 ± 2.31	87.41 ± 2.15	96.51 ± 0.33	86.25 ± 1.54
EK-ROP-Net	94.26 ± 0.80	98.54 ± 0.44	92.97 ± 1.29	90.91 ± 2.32	91.70 ± 1.52	97.84 ± 0.34	91.40 ± 1.21

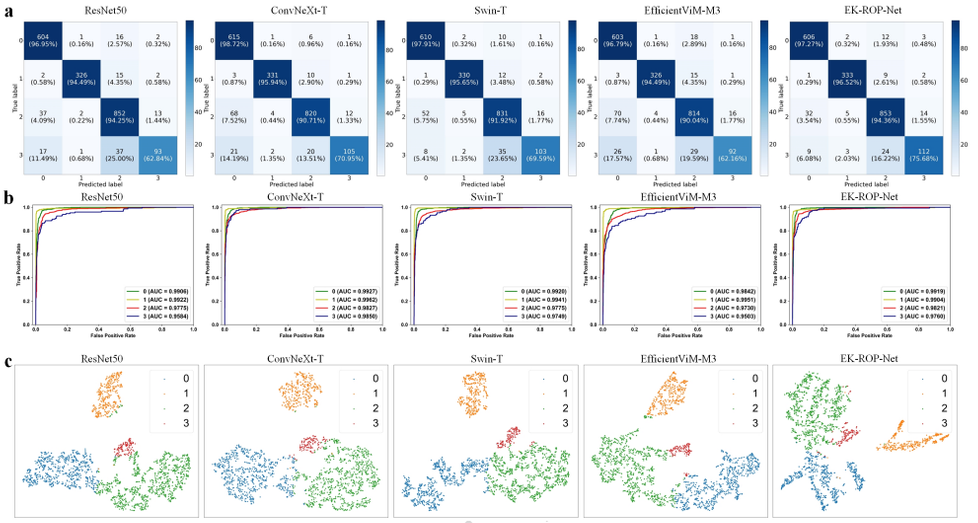


Fig. 8 Performance evaluation of AI models on the ROP Occurrence task for classifying four conditions: Normal (0), Laser-treated ROP (1), Any Stage of ROP (2), and A-ROP (3). The figure displays: (a) Confusion matrices (showing both image counts and normalized percentages), (b) Receiver Operating Characteristic (ROC) curves, and (c) t-SNE visualizations.

four-class ROP Occurrence task, all models exhibit high performance, with AUC scores generally exceeding 97%, which indicates strong discriminative ability. Notably, the vision-language model, EK-ROP-Net, outperforms the four image-only models on most evaluation metrics. As shown in Table 5, it achieves the highest accuracy (94.26 ± 0.80) and Kappa score (91.40 ± 1.21). The confusion matrices in Fig. 8a corroborate this, showing that EK-ROP-Net provides a more balanced classification, particularly in identifying the challenging A-ROP category (label 3).

Table 6 Model classification performances in ROP Severity task (%).

Method	Acc	AUC	Pre	Sen	F1	Spec	Kappa
ResNet50	83.73 ± 2.62	91.55 ± 2.80	85.34 ± 3.43	76.66 ± 5.06	79.46 ± 4.31	87.13 ± 2.38	65.72 ± 6.18
ConvNeXt-T	84.37 ± 1.12	93.56 ± 1.78	85.18 ± 2.92	78.06 ± 3.16	80.03 ± 1.93	87.73 ± 1.50	67.32 ± 3.11
Swin-T	86.51 ± 2.00	93.13 ± 1.85	86.92 ± 3.58	80.74 ± 4.43	83.33 ± 4.07	90.06 ± 1.36	72.52 ± 4.30
EfficientViM-M3	79.79 ± 1.46	89.29 ± 1.34	82.07 ± 2.12	66.74 ± 4.14	71.23 ± 3.86	84.56 ± 1.06	56.65 ± 3.16
EK-ROP-Net	87.70 ± 1.43	94.43 ± 0.76	88.65 ± 2.34	84.24 ± 2.04	86.16 ± 1.98	91.19 ± 0.99	75.49 ± 2.74

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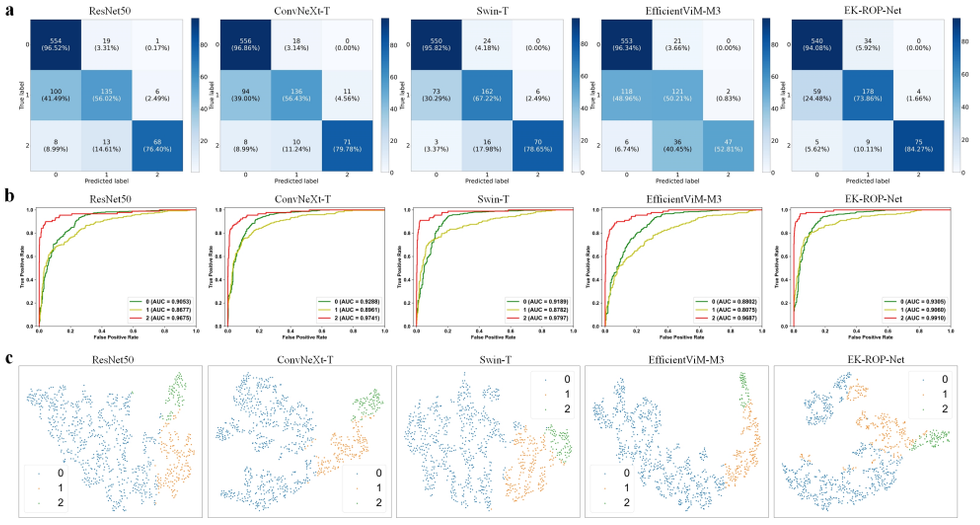


Fig. 9 Performance evaluation of AI models on the ROP Severity task for classifying three conditions: Mild ROP (0), Moderate ROP (1), Severe ROP (2). The figure displays: (a) Confusion matrices (showing both image counts and normalized percentages), (b) Receiver Operating Characteristic (ROC) curves, and (c) t-SNE visualizations.

The three-class ROP Severity task proves to be more challenging, a fact reflected by the lower overall metric scores across all models. This highlights the difficulty of fine-grained classification based on subtle pathological differences. Despite the increased difficulty, EK-ROP-Net again demonstrates its superiority, attaining the highest accuracy (87.70 ± 1.43) and AUC (94.43 ± 0.76) in Table 6. The confusion matrices in Fig 9a reveal that the image-only models exhibit significant confusion between adjacent classes (e.g., Mild and Moderate ROP), an issue that EK-ROP-Net mitigates more effectively, leading to a more reliable diagnosis.

In summary, the results validate that the ROP-VL dataset is a high-quality resource for AI model development in the ROP domain. The consistent leading performance of the EK-ROP-Net model underscores the significant advantage of integrating textual domain knowledge with visual data—a key feature enabled by our dataset. This indicates that the dataset is not only suitable for standard image classification but is also well-structured to facilitate the creation of more advanced, knowledge-aware vision-language models.

Multimodal vision–language applications. Beyond the image classification baselines reported here, ROP-VL is designed to support a wider range of vision–language tasks. The one-to-one text-image pairs are directly compatible with CLIP-style contrastive pretraining, enabling models to learn clinically grounded cross-modal representations. The structured descriptions in `disease_description.xlsx` serve as fixed class prototypes for zero-shot language-conditioned classification. The dataset also supports instruction fine-tuning of large vision–language models (e.g., LLaVA,

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BLIP), text-image retrieval for clinical decision support, and visual question answering. Researchers may adapt the $\pi_{\text{ROP-EK}}(\cdot)$ template to generate instruction-response pairs suitable for training or evaluating these architectures.

Limitations. Several limitations of the current dataset and technical validation should be acknowledged. (1) **Single-center, single-ethnicity**: ROP-VL was collected exclusively at SZEH and all subjects are of Chinese ethnicity, which may limit the generalizability of trained models to other populations or imaging environments. (2) **Class imbalance**: Severe ROP categories are substantially under-represented (Stage 4: 59 images, 2.9%; Stage 5: 30 images, 1.5%) relative to milder categories, posing challenges for model training and unbiased evaluation on rare categories. (3) **No external validation**: All validation is based on internal five-fold cross-validation; external validation on datasets from independent centers is a planned future direction. (4) **Imaging device heterogeneity**: Images from three different f undus imaging systems (RetCam, SW-8000, RS-B002) may exhibit inter-device variability in image characteristics. (5) **Subjective quality assessment**: Image quality screening was conducted through clinical visual inspection without automated quantitative metrics, which may limit the strict reproducibility of the inclusion criteria.

Data Availability

The dataset is available at https://doi.org/10.6084/m9.figshare.30143461.v1.

Code Availability

The source code for the baseline model training and testing will be available at https://github.com/SB-Chen/ROP-VL.

Author Contributions

Shaobin Chen, Tao Tan, Xinyu Zhao and Yue Sun: conceptualization, data collection, writing, reviewing; Zhenquan Wu and Guoming Zhang: data quality control, image classification; Shaobin Chen, Yue Sun and Jiaju Huang: methodology, formal analysis, supervision, reviewing; Yue Sun and Guoming Zhang: funding acquisition, project administration.

Competing Interests

The authors declare no competing interests.

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